

8.1 Analytical Techniques (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	8. Analysis (A Level Only)
Topic	8.1 Analytical Techniques (A Level Only)
Difficulty	Hard

Time allowed: 60

Score: /45

Percentage: /100

Question 1a

Compound **P** is a naturally occurring chemical found in strawberries, apples and Parmesan cheese.

The percentage by mass is carbon 58.82%, hydrogen 9.80% and oxygen 31.38%.

The mass spectrum of compound **P** is recorded in Fig. 1.1.

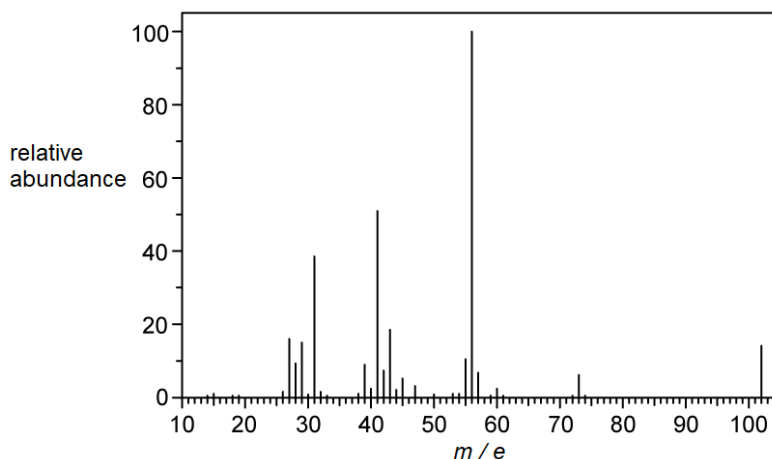


Fig. 1.1

Determine the molecular formula of compound **P**. Show your working.

[3 marks]

Question 1b

Table 1.1 shows the results of qualitative tests performed on compound **P**.

Table 1.1

Test	Observation
Addition of H ₂ O	Forms separate layers
Na ₂ CO ₃ (aq)	No visible change
2,4-DNPH	No visible change
Tollens' reagent	No visible change

Analyse the potential functional groups in compound **P**. Explain your answers.

[4 marks]

Question 1c

The carbon-13 (^{13}C) NMR spectrum of compound **P** is shown in Fig. 1.2.

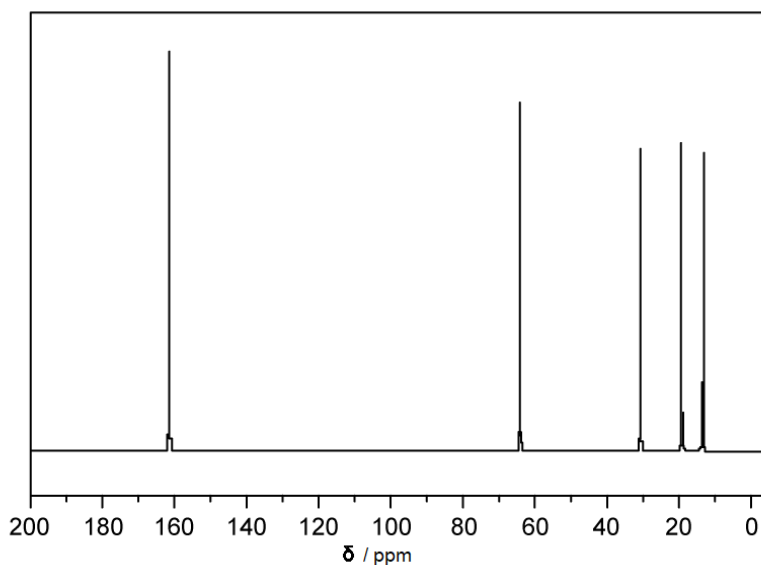
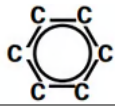


Fig. 1.2

Table 1.2

Hybridisation of the carbon atom	Environment of carbon atom	Example	Chemical shift range δ / ppm
sp^3	alkyl	CH_3- , CH_2- , $-\text{CH}-$, $>\text{C}-$	0 – 50
sp^3	next to alkene / arene	$-\text{C}-\text{C}=\text{C}-$, $-\text{C}-\text{Ar}$	25 – 50
sp^3	next to carbonyl / carboxyl	$\text{C}-\text{COR}$, $\text{C}-\text{O}_2\text{R}$	30 – 65
sp^3	next to halogen	$\text{C}-\text{X}$	30 – 60
sp^3	next to oxygen	$\text{C}-\text{O}$	50 – 70
sp^2	alkene or arene	$>\text{C}=\text{C}-$, 	110 – 160
sp^2	carboxyl	$\text{R}-\text{COOH}$, $\text{R}-\text{COOR}$	160 – 185
sp^2	carbonyl	$\text{R}-\text{CHO}$, $\text{R}-\text{CO}-\text{R}$	190 – 220
sp	nitrile	$\text{R}-\text{C}\equiv\text{N}$	100 – 125

Identify the functional group(s) present in compound **P** using your answer in (b) and information from Fig. 1.2 and Table 1.2. Explain your answer.

[3 marks]

Question 1d

The high-resolution **proton** NMR spectrum of compound **P** was recorded as shown in Fig. 1.3.

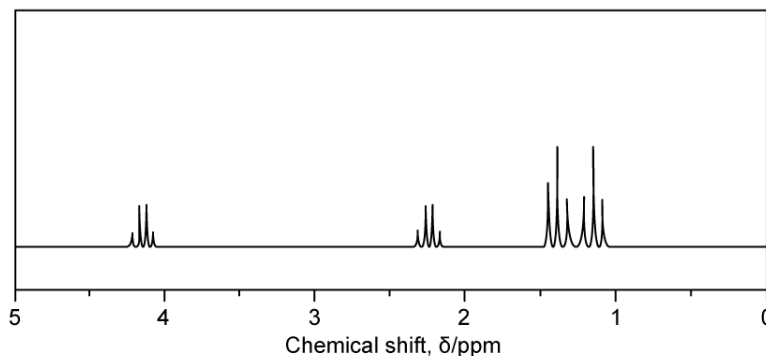


Fig. 1.3

Table 1.3

Environment of proton	Example	chemical shift range, δ / ppm
alkane	$-\text{CH}_3$, $-\text{CH}_2-$, $>\text{CH}-$	0.9 – 1.7
alkyl next to $\text{C}=\text{O}$	$\text{CH}_3-\text{C}=\text{O}$, $-\text{CH}_2-\text{C}=\text{O}$, $>\text{CH}-\text{C}=\text{O}$	2.2 – 3.0
alkyl next to aromatic ring	CH_3-Ar , $-\text{CH}_2-\text{Ar}$, $>\text{CH}-\text{Ar}$	2.3 – 3.0
alkyl next to electronegative atom	CH_3-O , $-\text{CH}_2-\text{O}$, $-\text{CH}_2-\text{Cl}$	3.2 – 4.0
attached to alkene	$=\text{CHR}$	4.5 – 6.0
attached to aromatic ring	$\text{H}-\text{Ar}$	6.0 – 9.0
aldehyde	$\text{HCO}-\text{R}$	9.3 – 10.5
alcohol	ROH	0.5 – 6.0
phenol	$\text{Ar}-\text{OH}$	4.5 – 7.0
carboxylic acid	RCOOH	9.0 – 13.0
alkyl amine	$\text{R}-\text{NH}-$	1.0 – 5.0
aryl amine	$\text{Ar}-\text{NH}_2$	3.0 – 6.0
amide	RCONHR	5.0 – 12.0

Suggest the structure of compound **P** using your answers to (a), (b) and (c) and information from Fig. 1.3 and Table 1.3. Explain your answer.

[3 marks]

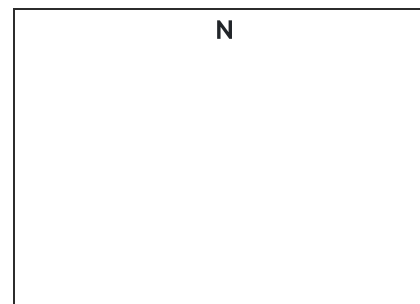
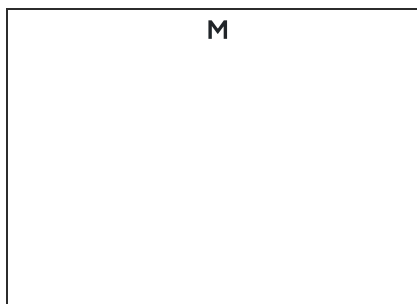
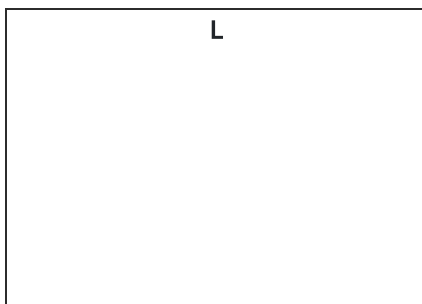
Question 2a

Three hydrocarbons, **L**, **M** and **N**, have the molecular formula C_8H_{10} . Information about the number of peaks seen in the carbon-13 (^{13}C) NMR spectrum of the three isomers is shown in Table 2.1.

Table 2.1

	Number of peaks
L	3
M	5
N	4

Suggest structures for compounds **L**, **M** and **N**.

**[3 marks]**

Question 2b

Complete Table 2.1 to give details of the proton NMR spectra for isomers **L**, **M** and **N**.

Table 2.1

	Number of peaks	Relative Peak area
L		
M		
N		

[3 marks]**Question 2c**

Explain which of the three isomers, **L**, **M** or **N** has the highest melting point.

[2 marks]

Question 3a

Gas chromatography is often connected to mass spectroscopy in order to analyse each compound as it exits the gas chromatography column. This combined technique is called GC-MS.

The gas chromatogram of an organic mixture is shown in Fig. 3.1. The stationary phase is a polar, high-boiling point liquid on a solid support.

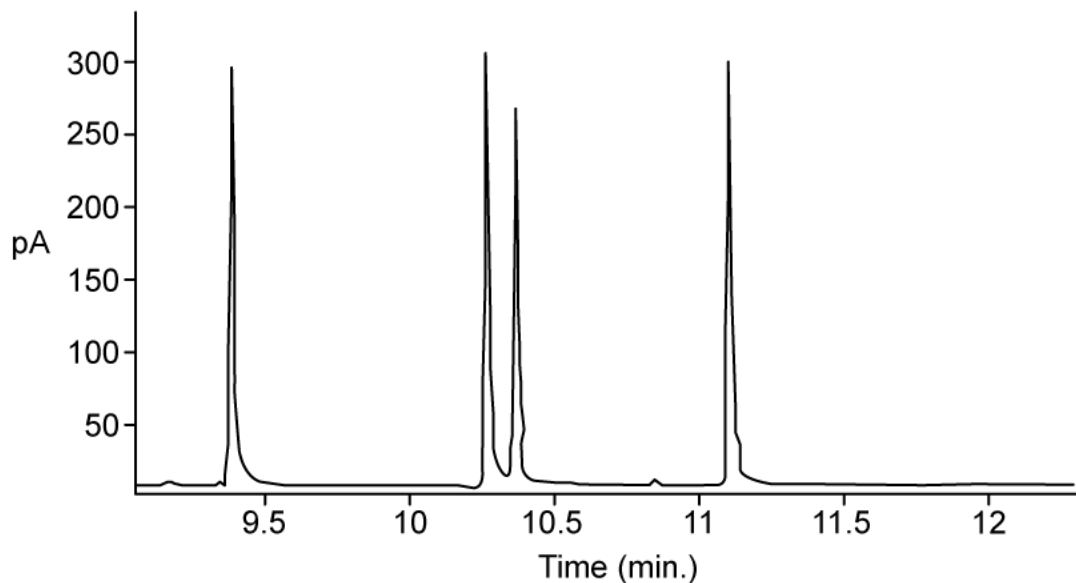


Fig. 3.1

- i)
Name an appropriate mobile phase that could be used to form this gas chromatogram.

[1]

- ii)
Identify the number of compounds present in this mixture and explain how to identify the most polar compound present.

[2]

[3 marks]

Question 3b

Two of the compounds in the organic mixture are hydrocarbons with similar retention times.

The mass spectra of the two compounds revealed that they have the same molecular formula but with subtle differences indicating structural isomerism is present.

The mass spectrum of one of the isomers is shown in Fig. 3.2.

Both isomers are tested with bromine water, which remains orange.

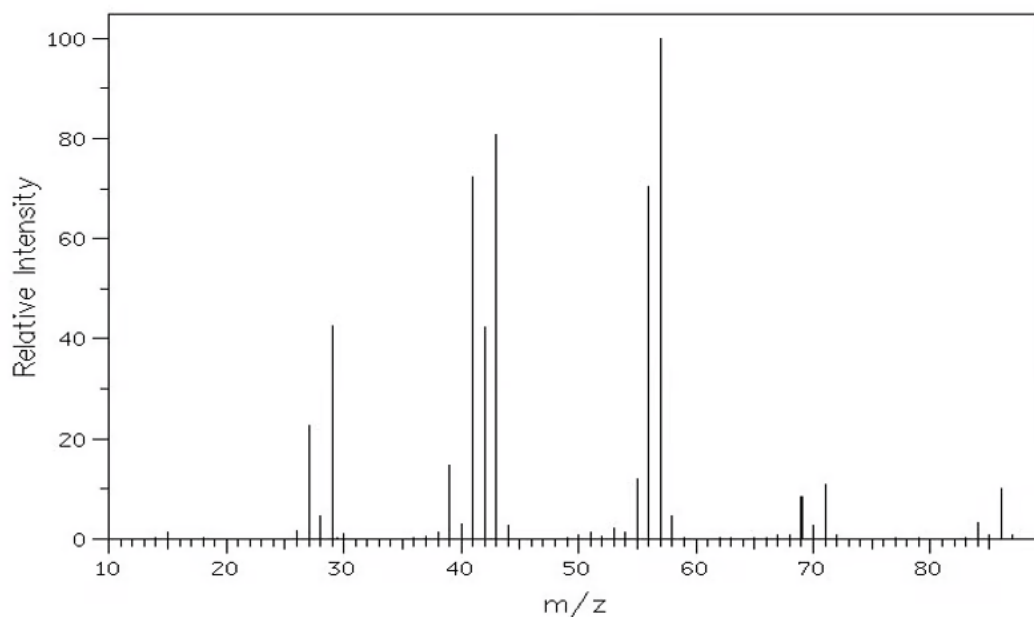


Fig. 3.2

- i)
Identify the molar mass and hence the molecular formula of the isomers.

molar mass

molecular formula

[2]

- ii)
Identify the fragments responsible for the following signals.

$m/e = 29$

$m/e = 43$

$m/e = 57$

[3]

iii)

Suggest the most likely structure of the isomer responsible for this spectrum.

[1]

[6 marks]

Question 3c

There is a small, additional peak on the far right of the mass spectrum.

i)

Explain the presence of the small peak at $m/e = 87$.

[1]

ii)

Suggest the type of molecules for which this would become more pronounced. Explain your answer.

[2]

[3 marks]

Question 3d

One other isomer present in the original mixture was 2,3-dimethylbutane.

- i)
Suggest **one** way in which the mass spectrum of 2,3-dimethylbutane will differ from Fig. 3.2 in part (b).

[1]

- ii)
Suggest an alternative spectroscopic technique for distinguishing between 2,3-dimethylbutane and your answer to (b).

[1]

- iii)
Predict which of the two isomers would have the longest retention time. Explain your answer.

[2]

[4 marks]

Question 4a

Dinitrobenzene has the molecular formula $C_6H_4N_2O_4$.

Draw the isomers of dinitrobenzene and name the type of isomerism.

[2 marks]

Question 4b

State the factors that determine the distance travelled by a spot in thin-layer chromatography.

[1 mark]

Question 4c

Fig. 4.1 shows the chromatogram with the spot for 1,4-dinitrobenzene.

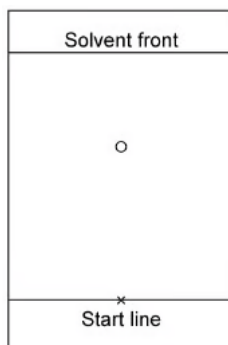


Fig. 4.1

Draw the expected position of the spot for 1,2-dinitrobenzene. Explain your answer.

[3 marks]

Question 4d

Explain what the student could do to reverse the relative positions of the 1,2-dinitrobenzene and 1,4-dinitrobenzene spots.

[2 marks]

Model Answers: Hard

1a

a) The molecular formula of the compound is:

- Mole ratio C : H : O = $\frac{58.82}{12.0} : \frac{9.80}{1.0} : \frac{31.38}{16.0}$
 $= 4.90 : 9.80 : 1.96$; [1 mark]
- Empirical formula = $C_5H_{10}O_2$; [1 mark]
- Molecular formula = $C_5H_{10}O_2 = (5 \times 12.0) + (10 \times 1.0) + (2 \times 16.0) = 102$
AND
Evidence of $m/e = 102$ peak in the spectrum; [1 mark]

ALTERNATIVE

- C: $\frac{102 \times \left(\frac{58.82}{100}\right)}{12.0} = 5.00$; [1 mark]
- H: $\frac{102 \times \left(\frac{9.80}{100}\right)}{1.0} = 10.0$

AND

- O: $\frac{102 \times \left(\frac{31.37}{100}\right)}{16.0} = 2$; [1 mark]
- Molecular formula $C_5H_{10}O_2$
AND
Evidence of $m/e = 102$ peak in the spectrum; [1 mark]

[Total: 3 marks]

- Use all the available information to solve the problem
 - This question gives you two routes to solving the problem if you know the percentage composition and the relative molecular mass
- Showing your steps not only is good practice but also helps you pick up on errors more easily

- You can then still receive some marks even if you make a mistake

1b

b) The functional groups in compound **P** and reasoning are:

- **P** does not contain a carboxylic acid
AND
The Na_2CO_3 (aq) test is negative; [1 mark]
- **P** does not contain an aldehyde or ketone
AND
The 2,4-DNP test is negative; [1 mark]
- **P** does not contain an aldehyde group
AND
The Tollens' test is negative; [1 mark]
- **P** could be an ester
OR
P could be an (insoluble) diol; [1 mark]

[Total: 4 marks]

- It is easier to say what the functional groups are not from the results of the tests
 - Negative results can be just as informative as positive results
- An ester is a good candidate as it has two oxygens and does not respond to any of the tests
 - Although esters are polar, they are not very soluble in water due to an inability to form hydrogen bonds
- Higher alcohols tend to be insoluble, but the presence of two -OH groups would increase the solubility
 - Without more information, it would be hard to say if it is a diol

1c

c) The functional group present in compound **P** is:

- Ester; [1 mark]

Explanation:

- The peak at $\delta = 162$ ppm corresponds to the $\text{C}=\text{O}$ of a carboxyl group; [1 mark]

- The peak at $\delta = 63$ ppm corresponds to a C-O; [1 mark]

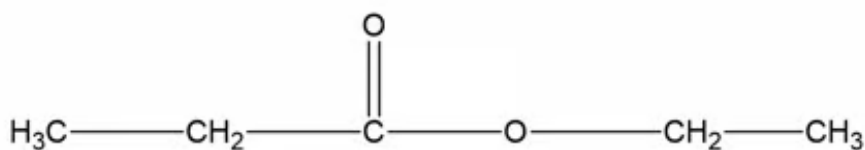
[Total: 3 marks]

- This is simply a case of matching up the spectrum with ranges given in Table 1.2
 - C-O is found at $\delta = 50 - 70$
 - C=O is found at $\delta = 160 - 185$
- An ester is consistent with the information given in parts (a) and (b)
 - Two oxygens are present
 - It is a functional group that does not react with Na_2CO_3 , 2,4-DNPH or Tollens' reagent

1d

d) The structure of compound **P** is:

- Correct structure; [1 mark]



Explanation:

Any **two** from:

- The spectrum shows two quartets and two triplets; [1 mark]
- A quartet and triplet in the same spectrum is indicative of an ethyl / CH_3CH_2 group; [1 mark]
- The spectrum shows two non-identical ethyl groups; [1 mark]

[Total: 3 marks]

- There are several isomeric esters possible with this molecular formula
 - It is the splitting patterns in the proton NMR that allow you to deduce the specific isomer
- Tell-tale patterns like doublets, triplets and quartets enable you to determine neighbouring protons and hence the structure
- Make sure you know where characteristic functional groups are found in the

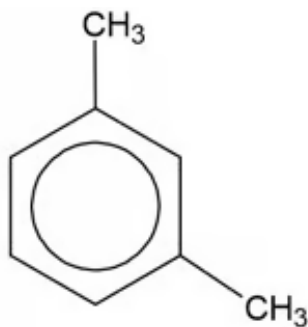
spectrum as you can also be asked to identify particular peaks in ^{13}C or ^1H NMR spectra

2a

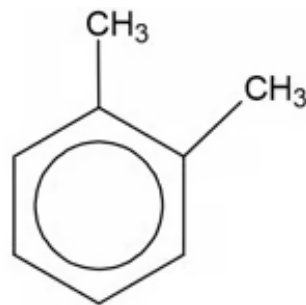
a) The structures of **L**, **M** and **N** are:



L



M

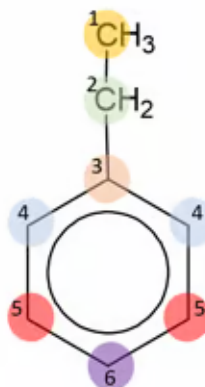


N

- Correct structure of **L**; [1 mark]
- Correct structure of **M**; [1 mark]
- Correct structure of **N**; [1 mark]

[Total: 3 marks]

- The low carbon-to-hydrogen ratio tells you it must be highly unsaturated or aromatic
 - Therefore, the benzene ring, C_6H_6 is a good starting point
- That leaves two carbons either together or separately joined on the ring
- Placing the two remaining carbons together would leave the ring as C_6H_5 , so the five remaining hydrogens would make it an ethyl group side group, $-\text{CH}_2\text{CH}_3$
 - This option can be eliminated as it would have 6 unique environments due to symmetry:



- Placing the two remaining carbons apart, but joined on the ring separately would leave the ring as C_6H_4 , making the two side groups both methyls, $-CH_3$
 - Next, assign the unique environments to the remaining structures, which have the formula $C_6H_4(CH_3)_2$
 - A line of symmetry through the structures helps to identify which are unique carbon environments:

L	M	N
3 unique environments	5 unique environments	4 unique environments

2b

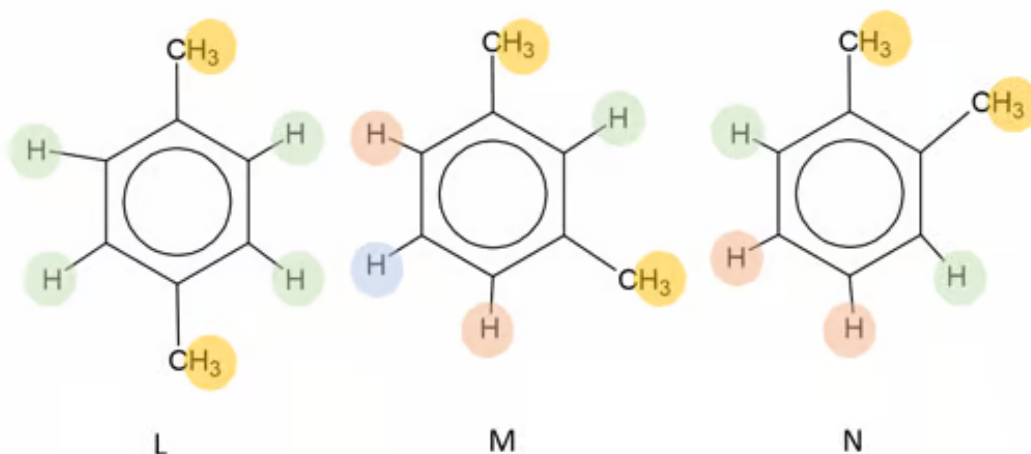
b) The completed table of proton NMR for L, M and N is:

	Number of peaks	Relative Peak area	
L	2	3:2	; [1 mark]
M	4	6:1:2:1	; [1 mark]
N	3	3:1:1	; [1 mark]

[Total: 3 marks]

- Having deduced the structures of **L**, **M** and **N** in part (a), you can now assign proton environments to the structures

- The protons have been added to the skeletal drawings to make them easier to see:



- **L** has two unique proton environments
 - The CH_3 groups which account for 6 protons in one environment
 - The benzene C-H accounts for 4 protons in one environment
 - This gives an overall ratio 6 : 4 which simplifies to 3 : 2
- **M** has four unique proton environments
 - The CH_3 groups which account for 6 protons in one environment
 - The benzene C-H accounts for 4 protons in three environments with a ratio of 1 : 2 : 1
 - This gives an overall ratio of 6 : 1 : 2 : 1
- **N** has three unique proton environments
 - The CH_3 groups which account for 6 protons in one environment
 - The benzene C-H accounts for 4 protons in two environments with a ratio of 2 : 2
 - This gives an overall ratio of 6 : 2 : 2, which simplifies to 3 : 1 : 1

2c

c) The isomer with the highest melting point is:

- Isomer L; [1 mark]

Explanation:

- Linear molecule packs more easily / closely
AND

Have the strongest intermolecular forces / induced dipole-dipole attractions (as a result of this packing); [1 mark]

[2 marks]

- Molecular packing is the main criterion for determining melting point order
- 1,4- isomers / para-isomers (those with the side groups directly opposite) are symmetrical and fit more easily into a regular repeating structure (lattice)
 - This is similar to the idea of intermolecular forces in branched / non-branched compounds.

3a

a)

i) An appropriate mobile phase would be:

- Helium / argon / nitrogen; [1 mark]

ii) The number of compounds present is:

- 4 / four; [1 mark]

How to identify the most polar compound:

- The most polar compound has the longest retention time

OR

The most polar compound has a retention time of 11.1 minutes; [1 mark]

[Total: 3 marks]

- You need to be able to name examples of inert gases, not just say 'an inert gas'
- A polar stationary phase will attract a polar molecule more
 - This will slow the progress of that compound through the gas chromatography column

3b

b)

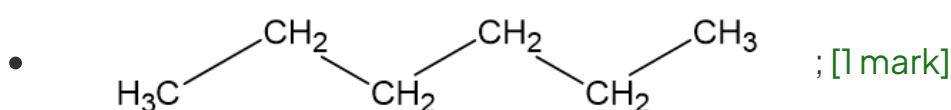
i) The molar mass and molecular formulae of the isomers are:

- Molar mass = 86 g mol^{-1} ; [1 mark]
- Molecular formula = C_6H_{14} ; [1 mark]

ii) The fragments responsible are:

- $m/z = 29 = \text{C}_2\text{H}_5^+$; [1 mark]
- $m/z = 43 = \text{C}_3\text{H}_7^+$; [1 mark]
- $m/z = 57 = \text{C}_4\text{H}_9^+$; [1 mark]

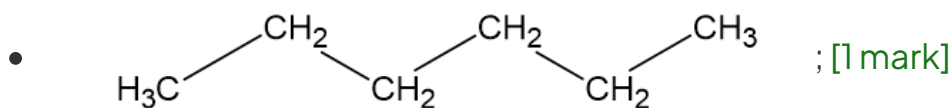
iii) The structure of the isomer most likely to give this spectrum is:



[Total: 6 marks]

- The molecular ion peak is equal to the molar mass, assuming single positive charges only
- The question tells you that:
 - The molecules are hydrocarbons
 - The molecules are saturated / contain only hydrogen and carbon with single bonds present since bromine water does not decolourise
- Using this information and the mass spectrum, you can determine the molecular formula:
 - The mass spectrum gives a molecular ion peak at $m/e = 86 \text{ g mol}^{-1}$
 - To satisfy 86 g mol^{-1}
 - 7 carbon atoms would require 2 hydrogen atoms giving an unrealistic molecular formula of C_7H_2 - which also does not agree with the bromine water test
 - 6 carbon atoms would require 14 hydrogen atoms giving a molecular formula of C_6H_{14} - which agrees with the bromine water test
 - 5 carbons would require 26 hydrogen atoms giving an unrealistic molecular formula of C_5H_{26}
- To gain the marks, the fragmentation answers all need to have positive charges
 - **Remember:** Your previous answers and information given in this question mean that all the fragments contain only carbon and hydrogen

- The fragment at $m/e = 57$ corresponds to a chain of 4 carbon atoms with 9 hydrogen atoms
 - This indicates that the overall molecule is straight-chain hexane, as no branched isomers would produce a C_4H_9^+ fragment



[Total: 6 marks]

- The molecular ion peak is equal to the molar mass, assuming single positive charges only
- The question tells you that:
 - The molecules are hydrocarbons
 - The molecules are saturated / contain only hydrogen and carbon with single bonds present since bromine water does not decolourise
- Using this information and the mass spectrum, you can determine the molecular formula:
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- The fragment at $m/e = 57$ corresponds to a chain of 4 carbon atoms with 9 hydrogen atoms
 - This indicates that the overall molecule is straight-chain hexane, as no branched isomers would produce a C_4H_9^+ fragment

3c

c)

i) The peak at $m/e = 87$ can be explained by:

- The presence of carbon-13 / ^{13}C in the molecule; [1 mark]

ii) The types of molecules where the additional peak becomes more pronounced are:

- Longer carbon chains / compounds with more carbon (atoms); [1 mark]

Explanation:

- As the number of carbons in the molecule / compound / chain increases
AND
The probability that one is carbon-13 / ^{13}C increases; [1 mark]

[Total: 3 marks]

- You need to be able to identify that a very small peak at $M+1$ is **not** the molecular ion peak
 - The molecular ion peak is the mass of the most common isotope, i.e. assumes that all carbon atoms are carbon-12 / ^{12}C)
- The probability of carbon-13 / ^{13}C increases with the number of carbons
 - Therefore, the peak size will be greater with more carbons in the molecule / compound / chain as the probability of atoms being carbon-13 / ^{13}C increases

3d

d)

i) One way in which the spectra for 2,3-dimethylbutane would be different is:

Any **one** from:

- No peak / minimal peak at $m/e = 57$ (as C_4H_9^+ fragment is not viable); [1 mark]
- No peak / minimal peak at $m/e = 15$ peak (as CH_3^+ is less likely to form); [1 mark]
- Peak at $m/e = 27$ (for C_2H_3^+); [1 mark]

ii) An alternative technique for distinguishing between the two isomers is:

- C-13 NMR
OR
Proton NMR; [1 mark]

iii) The isomer with the longest retention time would be:

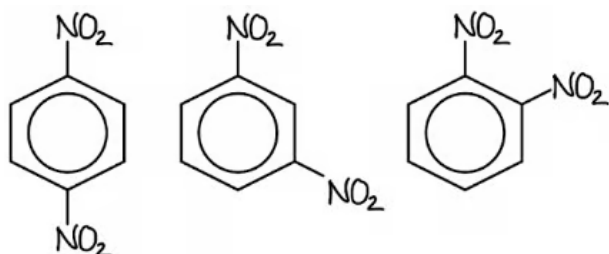
- 2,3-dimethylbutane; [1 mark]
- As the branching would cause it to 'snag' on the stationary phase/ more loose ends to get caught in the stationary phase; [1 mark]

[Total: 4 marks]

- The fragments come from the structure breaking apart in the initial stages of mass spectrometry and then an ionised fragment making it through
 - This means that the fragments have to be jigsaw pieces that would be able to fit together to make the molecule
 - The jigsaw pieces for the branched molecule are different from the straight-chained one
- It is much easier to use NMR to distinguish between structural isomers!
- The retention time depends on the affinity for each of the phases
 - The stationary phase is a high-boiling point liquid on a solid support
 - Branched molecules are more likely to get stuck and, therefore, take longer to pass through

4a

a) The isomers of dinitrobenzene are:



- All **three** structures correct; [1 mark]
- (This is) positional / position isomerism; [1 mark]

[Total: 2 marks]

- It's surprising how many errors are made drawing di-substituted isomers of benzene
- **Tip:** Place one functional group at the top (carbon-1) and do **not** move that functional group at all
 - Place the other functional group on the adjacent carbon (carbon-2)
 - Move the second functional group around the benzene ring one carbon at a

time until you are opposite the first functional group (carbon-4)

4b

b) The factors that determine the distance travelled by a spot in thin-layer chromatography are:

Either of the following alternatives:

Alternative 1

- The solubility of the spot in the mobile phase / liquid / solvent

AND

The retention of the compound by stationary phase / solid / column; [1 mark]

OR

Alternative 2

- The affinity of the compound with the different phases / mobile and stationary phases; [1 mark]

[Total: 1 mark]

- The movement of any compound during chromatography is dependent on its interactions with the mobile and stationary phases
 - Solubility and retention
- These interactions are influenced by underlying points of chemistry such as the polarity of the compound, the solvent and the stationary phase medium

4c

c) The expected position of the spot for 1,2-dinitrobenzene is:



- Spot correctly placed below the one already on the chromatogram; [1 mark]

Explanation:

- The silica gel / solid phase is polar

OR

The hexane / solvent / mobile phase is non-polar; [1 mark]

- 1,2-dinitrobenzene is more polar

AND

1,2-dinitrobenzene has a greater affinity for the stationary phase / has a lower affinity for the mobile phase

AND

Therefore, 1,2-dinitrobenzene does not travel as far; [1 mark]

[Total: 3 marks]

- Both compounds in question are dinitrobenzene isomers so the difference on the chromatogram must be due to something about their actual structures
 - 1,2-dinitrobenzene has both nitro groups next to each other and is therefore more polar
 - This means that it is more attracted to the solid phase / silica gel
 - 1,4-dinitrobenzene has both nitro groups on opposite sides of the benzene ring and is therefore less polar
 - This means that it is more soluble in the mobile phase / hexane

4d

e) To reverse the relative positions of the 1,2-dinitrobenzene and 1,4-dinitrobenzene spots, the student could:

- Replace the hexane with a polar solvent / use a polar solvent; [1 mark]

Explanation

- 1,2-dinitrobenzene will be more attracted to / more soluble in / have a stronger affinity with the solvent (than before)

AND

Therefore, the 1,2-dinitrobenzene spot will travel further; [1 mark]

[Total: 2 marks]

- This question tests your understanding of the interactions of polar compounds with polar and non-polar solvents
- Polar compounds are more soluble in polar solvents
 - This means that they have a greater affinity with a polar solvent and, therefore, travel further